PostgreSQL vs MongoDB Pattern 1 vs MongoDB Pattern 2

Final experimental results

1. Experimental scope and measurement protocol

Pattern 1 (P1) is sample-centred: each sample embeds a large totalProteomeMeasurements[] array. Pattern 2 (P2) keeps sample metadata separate and stores each protein measurement as an independent document in protein_measurements. The original P1 database remains unchanged, allowing direct A/B comparison.

Migration checks: 304,911 P1 embedded measurements = 304,911 P2 measurement documents; CTRL_BASAL_1 contains 6,376 measurements in both representations. phosphopeptide_summaries is structurally unchanged; equivalent indexes were mirrored before control-query comparisons.

Measurement protocol: cache warm-up uses five executions before formal measurement, excluded from the reported statistics. Each reported configuration contains ten measured values. MongoDB timing uses Compass Explain; medians, means, ranges and sample standard deviations describe the recorded execution times.

Control queries: Q3, Q7 and optimized Q8 use the unchanged phosphopeptide summary collection. One shared set of measurements is reported for P1/P2 where indicated (S), with equivalent indexes and execution paths checked in both patterns.

2. Overall benchmark summary

Median execution time in milliseconds. Statistics are calculated from the supplied recorded values.

Query	PostgreSQL	P1 baseline	P1 optimized	P2
Q1 Condition-Level Protein Abundance Aggregation	366.525	360	-	564.5
Q2 Defective Phosphosite-Protein Measurement Lookup	15.28	541	293.5	5
Q3 Gene-Level Dysregulated Phosphosite Aggregation	13.0945	10 S	-	10 S
Q4 Selective Akt1 Dataset-Level Lookup	1.7505	176.5	118.5	1
Q5 Akt1 Retrieval in Insulin-Resistant INS Samples	0.46	112.5	105	1
Q6 INS vs BASAL Akt1 Aggregation	2.089	200.5	138	2
Q7 Gene-Specific Phosphopeptide Annotation Retrieval	1.9965	3 S	-	3 S
Q8 Model-Specific Dysregulation Retrieval	86.13	65.5	9.5 S	9.5 S

S = one shared P1/P2 measurement set shown in both applicable columns. - = no separate configuration reported.

Result pattern: P2 is substantially faster than optimized P1 for selective protein queries, while Q1 is slower in P2. Q3, Q7 and Q8 use the unchanged summary collection and serve as controls for the schema comparison.

3. Q1 - Condition-Level Protein Abundance Aggregation

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	366.525	357.2936	327.085-372.003	17.9
MongoDB P1 baseline	10	360	362.6	341-386	15.5
MongoDB P2	10	564.5	569.9	511-688	47.9

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 327.085, 370.158, 333.364, 368.231, 335.329, 369.260, 364.819, 361.226, 372.003, 371.461

MongoDB P1 baseline runs (ms): 374, 376, 386, 379, 366, 350, 341, 354, 351, 349

MongoDB P2 runs (ms): 511, 520, 552, 579, 578, 561, 581, 565, 688, 564

Explain evidence: P2 performed a COLLSCAN over all 304,911 protein_measurements documents. A compound index over condition fields, geneSymbol, uniprotId and abundance was tested, but MongoDB still selected COLLSCAN because Q1 has no selective initial filter and must aggregate the entire collection.

Interpretation: Medians are 366.525 ms for PostgreSQL, 360 ms for P1 and 564.5 ms for P2. P1 is better aligned with this broad aggregation. Pre-aggregated/materialized summaries are a potential optimization for frequent whole-dataset summaries.

4. Q2 - Defective Phosphosite-Protein Measurement Lookup

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	15.28	14.6035	9.322-20.927	3.5
MongoDB P1 baseline	10	541	539.7	491-589	27.9
MongoDB P1 optimized	10	293.5	286.8	239-313	23.8
MongoDB P2	10	5	6.1	4-12	2.7

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 11.184, 16.788, 15.812, 15.098, 20.927, 9.322, 15.462, 17.112, 10.389, 13.941

MongoDB P1 baseline runs (ms): 491, 562, 530, 552, 522, 555, 525, 517, 554, 589

MongoDB P1 optimized runs (ms): 310, 313, 256, 291, 277, 239, 305, 296, 284, 297

MongoDB P2 runs (ms): 4, 5, 9, 12, 4, 4, 8, 4, 5, 6

Explain evidence: After fair indexing, the outer phosphopeptide match examined 5 documents and 5 index keys using insulinRegulationStatus_1. P2 also uses proteinAnalyteId_1 for direct access to measurement documents rather than filtering an embedded measurement array in each sample.

Interpretation: The optimized P1 median is 293.5 ms versus 5 ms for P2, approximately 58.7 times faster in P2. P2 is also faster than PostgreSQL (15.28 ms) in this workload. The execution plans support a structural explanation beyond indexing alone.

5. Q3 - Gene-Level Dysregulated Phosphosite Aggregation

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	13.0945	12.9110	10.394-15.363	1.4
MongoDB P1 / P2 shared	10	10	10.9	9-18	2.8

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 11.580, 12.313, 12.972, 13.486, 10.394, 12.390, 15.363, 13.705, 13.690, 13.217

MongoDB P1 / P2 shared runs (ms): 9, 10, 9, 13, 10, 18, 12, 9, 10, 9

Explain evidence: P1 and P2 both used insulinRegulationStatus_1, examined 1,663 documents and 1,664 index keys, and followed the same IXSCAN -> FETCH -> PROJECTION path.

Interpretation: The shared MongoDB measurements have a 10 ms median; PostgreSQL has a 13.0945 ms median. This query operates entirely on phosphopeptide_summaries, whose structure and indexed access path are unchanged between P1 and P2.

6. Q4 - Selective Akt1 Dataset-Level Lookup

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	1.7505	1.7914	0.772-5.175	1.3
MongoDB P1 baseline	10	176.5	180.9	163-209	17.3
MongoDB P1 optimized	10	118.5	118	90-154	18.7
MongoDB P2	10	1	1.1	0-2	0.6

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 0.772, 2.055, 1.803, 0.829, 5.175, 1.698, 0.834, 2.084, 0.814, 1.850

MongoDB P1 baseline runs (ms): 163, 197, 209, 169, 206, 179, 182, 174, 164, 166

MongoDB P1 optimized runs (ms): 114, 120, 90, 154, 90, 117, 116, 123, 124, 132

MongoDB P2 runs (ms): 1, 2, 2, 1, 1, 1, 0, 1, 1, 1

A MongoDB 0 ms reading reflects the displayed millisecond resolution and does not imply zero execution time.

Explain evidence: Both optimized P1 and P2 locate 48 Akt1 matches with 48 index keys. P1 uses a multikey index but must still fetch large sample documents and filter totalProteomeMeasurements[]. P2 uses a non-multikey geneSymbol_1 index and directly fetches the 48 matching measurement documents.

Interpretation: P1 optimization reduces its median from 176.5 to 118.5 ms. P2 reaches a 1 ms median, corresponding to approximately 118.5 times faster execution than optimized P1. The execution plans are consistent with the remaining array-processing cost in P1.

7. Q5 - Akt1 Retrieval in Insulin-Resistant INS Samples

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	0.46	0.5334	0.343-1.072	0.2
MongoDB P1 baseline	10	112.5	117	101-139	11.7
MongoDB P1 optimized	10	105	104	79-140	18.2
MongoDB P2	10	1	1	0-2	0.7

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 0.623, 0.459, 0.435, 0.578, 0.461, 1.072, 0.438, 0.414, 0.511, 0.343

MongoDB P1 baseline runs (ms): 134, 110, 139, 113, 112, 118, 101, 110, 111, 122

MongoDB P1 optimized runs (ms): 83, 111, 91, 100, 95, 113, 140, 118, 110, 79

MongoDB P2 runs (ms): 0, 1, 1, 1, 1, 2, 0, 2, 1, 1

A MongoDB 0 ms reading reflects the displayed millisecond resolution and does not imply zero execution time.

Query optimization: the P1 optimization filters before unwind to reduce unnecessary array expansion.

Interpretation: P1 optimization reduces the median from 112.5 to 105 ms. P2 has a 1 ms median, approximately 105.0 times faster than optimized P1. PostgreSQL has a 0.46 ms median. The P1/P2 contrast remains consistent with the cost of inspecting embedded arrays.

8. Q6 - INS vs BASAL Akt1 Aggregation

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	2.089	1.9984	1.337-2.588	0.5
MongoDB P1 baseline	10	200.5	201.5	177-230	13.8
MongoDB P1 optimized	10	138	136.6	112-155	15.6
MongoDB P2	10	2	2	1-3	0.8

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 1.337, 2.278, 2.111, 1.511, 2.588, 2.067, 1.463, 2.572, 1.784, 2.273

MongoDB P1 baseline runs (ms): 196, 200, 204, 191, 177, 201, 204, 199, 230, 213

MongoDB P1 optimized runs (ms): 112, 149, 131, 131, 155, 122, 119, 145, 149, 153

MongoDB P2 runs (ms): 1, 2, 2, 3, 1, 3, 2, 1, 3, 2

Query optimization: the P1 optimization avoids full-array unwind; P2 aggregates directly over relevant measurement documents.

Interpretation: Optimized P1 has a 138 ms median, compared with 2 ms for P2 (about 69 times faster). PostgreSQL has a 2.089 ms median from ten recorded values. Its small difference from P2 should be interpreted with the different timing resolutions in mind.

9. Q7 - Gene-Specific Phosphopeptide Annotation Retrieval

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	1.9965	2.612	1.781-4.768	1.1
MongoDB P1 / P2 shared	10	3	3.1	3-4	0.3

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 2.008, 1.846, 1.951, 4.304, 2.084, 1.985, 3.487, 1.906, 4.768, 1.781

MongoDB P1 / P2 shared runs (ms): 3, 3, 3, 3, 4, 3, 3, 3, 3, 3

Explain evidence: With the same geneSymbol_1 index, both patterns examined 131 documents and 131 index keys and returned the same 131 documents.

Interpretation: The shared MongoDB measurements have a 3 ms median. PostgreSQL has a 1.9965 ms median. Equivalent geneSymbol indexes produce matching examined-record counts and result counts in P1 and P2.

10. Q8 - Model-Specific Dysregulation Retrieval

System / configuration	n	Median	Mean	Range (ms)	SD
PostgreSQL	10	86.13	87.4783	84.06-93.297	3.3
MongoDB P1 baseline	10	65.5	65.3	58-75	5.7
MongoDB P1 / P2 optimized shared	10	9.5	9.6	8-11	1.0

All times and SD are in ms; SD is the sample standard deviation.

PostgreSQL runs (ms): 84.060, 86.743, 84.669, 90.430, 86.203, 85.521, 85.462, 93.297, 86.057, 92.341

MongoDB P1 baseline runs (ms): 59, 70, 71, 67, 58, 59, 75, 63, 64, 67

MongoDB P1 / P2 optimized shared runs (ms): 10, 9, 9, 11, 10, 9, 10, 9, 11, 8

Explain evidence: With equivalent defects.DEX_1 and emergent.DEX_1 indexes, P1 and P2 both examined 494 documents and 494 index keys using two IXSCAN branches under OR, returning 50 documents.

Interpretation: The P1 baseline median is 65.5 ms; the shared optimized MongoDB median is 9.5 ms. PostgreSQL has a 86.13 ms median. The execution plans support the importance of predicate and index design.

11. Cross-query interpretation

Measurement-centred documents benefit selective protein access (Q2, Q4-Q6), while broad whole-dataset aggregation (Q1) is a trade-off. Q3, Q7 and Q8 exercise the unchanged phosphopeptide summary path and serve as controls with equivalent indexes. The results support workload-aware schema design: measurement-centred documents suit selective retrieval and aggregation, while broad summaries may benefit from sample-centred storage or materialized summaries.